Students Academic Performance: Does Parents' Education Matter?

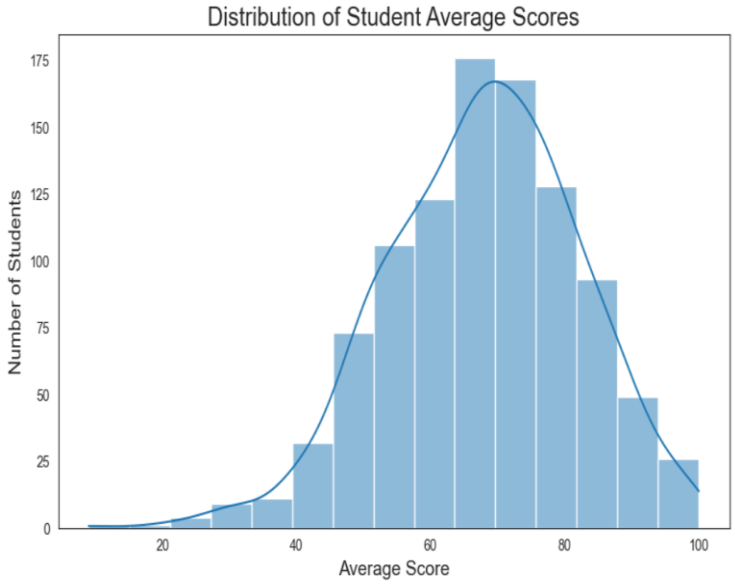
DATA VISUALIZATION

Target audience: educators, school administrators, and education policymakers interested in understanding and reducing academic performance gaps.

Hussein Shamas

Data Visualization

Individual Assignment



- Most scores fall between
- 60 - 80.
- Few very low (<40) or very high (>95).
- Overall balanced spread.

Students Academic Performance: To what extent does parental education level influence students' academic performance across subjects, and how does this interact with gender and test preparation?

Student name: Hussein Shamas
Student ID: 946916
Student e-mail address: H.shamas@unimib.campus.gmail.com

About Data

Initial considerations:

•The dataset provides detailed academic performance (math, reading, writing) along with demographic and family information for students.

I wanted a dataset that could help answer whether **parents' education level affects student performance**.

Issues faced while searching for data:

- Many publicly available datasets either lacked parental education info or had incomplete score data.
- Some datasets were focused on specific regions or grades, not providing a generalizable sample.

Chosen dataset(s):

•**"Students Performance in Exams"** (Kaggle) – contains 1000+ students, includes math, reading, writing scores, gender, test prep course, and parental education level. Link: StudentsPerformance

License:

•Shared under **Kaggle's terms of use**, allowing non-commercial research and academic work.

•This means I can analyze, visualize, and present findings for educational purposes but cannot redistribute the raw dataset publicly.

Data quality & completeness concerns:

•Dataset is **mostly clean**, but potential issues include:

- Some parental education categories have fewer students (imbalanced groups).
- Cross-sectional survey – cannot analyze trends over time.
- Scores are standardized but may have minor input errors (unlikely, given Kaggle verification).

About Data

Anticipated data cleaning operations:

- Renaming columns for clarity.
- Creating derived variables like average score or total score.
- Converting categorical columns (e.g parental level of education) to consistent naming.
- Checking for missing values (none in this dataset, but verified programmatically).

Methodology

Data Collection:
- Dataset obtained from **Kaggle** ("Students Performance in Exams"), downloaded manually.
- No scraping or APIs were used, as the dataset is fully provided in CSV format.

Data Processing & Cleaning:
- Processed using **Python** in **Jupyter Notebook**.
- Key libraries: **pandas** (data handling), **numpy** (numeric operations), **matplotlib & seaborn** (visualization).
- Checked for missing values, corrected inconsistent column names, and verified data types.

Data Transformation:
- Created derived variables:
    - Average score = mean of math, reading, and writing scores per student
    - Total_score = sum of math, reading, and writing scores
- Aggregated data by **parental education level** for analysis.
- Encoded categorical variables consistently for plotting and statistical analysis.

- **Use of AI Tools:**
- **ChatGPT** used for:
    - Suggestions for data cleaning and analysis steps.
- All analysis decisions and interpretation were done independently.

Methodology

Analytical Techniques:

- **Descriptive statistics:** mean, median, standard deviation of scores per parental education group.
- **Comparative analysis:** boxplots to compare distributions across groups.
- **Subject-level comparison:** math, reading, writing scores analyzed separately to identify patterns
- **Gender & test preparation interactions**: additional stratified analysis with visualizations.

Reproducibility:

- All code shared in **Jupyter Notebook**.
- Cleaned dataset exported as students_performance_clean.csv.
- All steps documented and can be replicated using the provided code and Kaggle dataset.
- All was documented on my **Github**: RepoLink

Visual Design Choices

- A neutral and minimal color palette was chosen to keep the focus on data comparison rather than decoration.

- Consistent colors were used across all charts to represent parental education levels, ensuring visual coherence across the project.

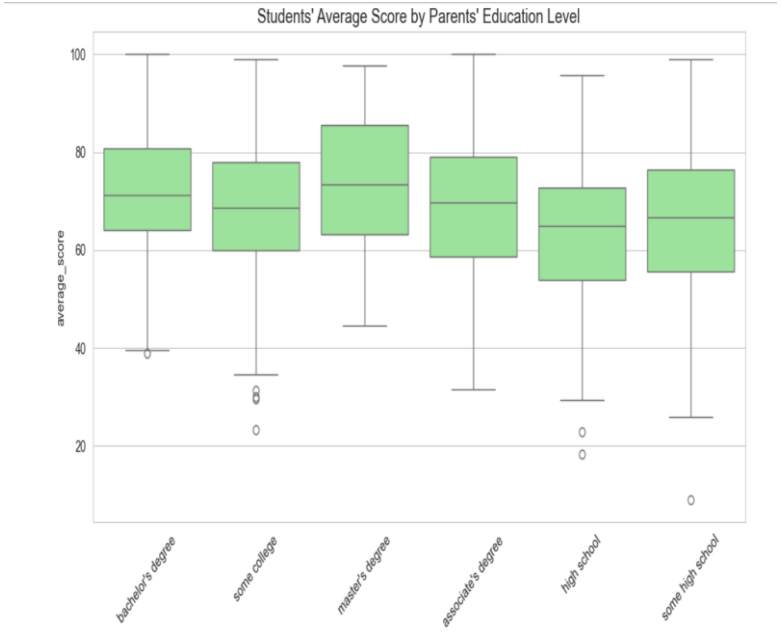
- Standard presentation fonts were used consistently across all slides to ensure readability and visual clarity.

- No major changes to palette or typography were required during the project, as the initial design choices already supported clarity and interpretability.

Insights from the Data

Main Insights:

- **Parental education matters:** Students whose parents have higher education (bachelor's or master's degree) consistently show **higher average scores** across math, reading, and writing.
- **Score ranges:**
  - Minimum average score: ~50
  - Maximum average score: ~100
  - Average overall score: ~70-75
- **Outliers:** A few students with very low scores (below 40) and very high scores (above 95) exist, but distribution is generally normal.
- **Subject differences:** Reading and writing scores show a stronger correlation with parental education than math.
- **Gender trends:** Girls slightly outperform boys in reading and writing, while math scores are similar across genders.
- **Test preparation effect:** Completing a test preparation course boosts scores across all parental education groups.



- Higher parental education → higher scores.
- Clear gap between groups.
- Pattern is consistent.

Insights from the Data

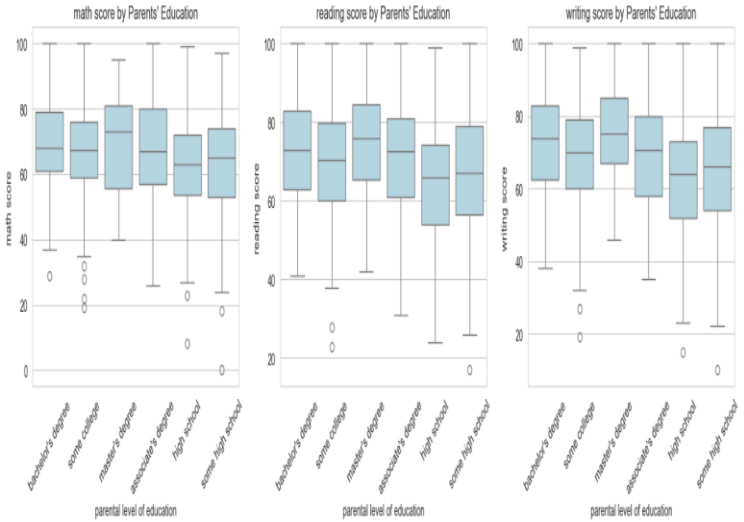
Analysis Operations Performed:

- **Descriptive statistics:** mean, median, min, max, standard deviation per parental education group.

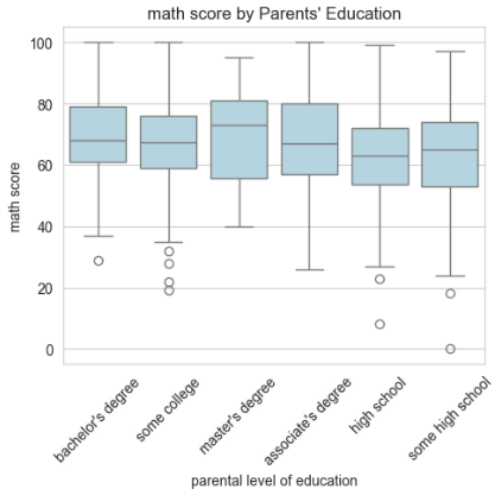
- **Comparative analysis:** boxplots comparing average scores across parental education categories.

- **Subject-level comparison:** separate boxplots for math, reading, and writing scores.

- **Stratified analysis:** additional boxplots by gender and test preparation course.

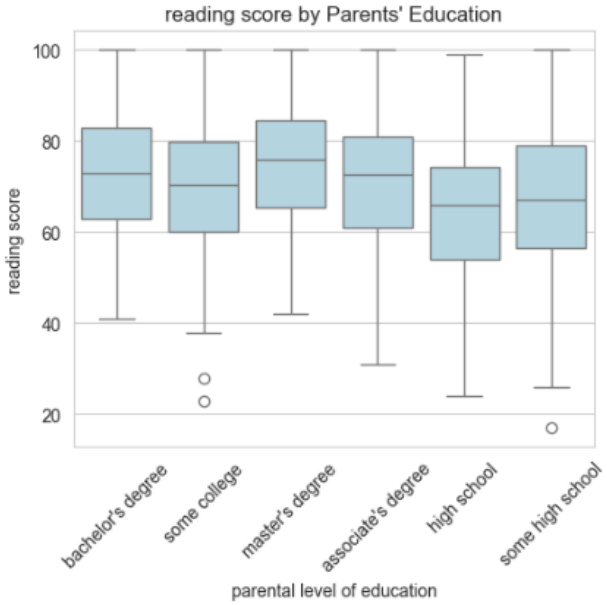


Math Score



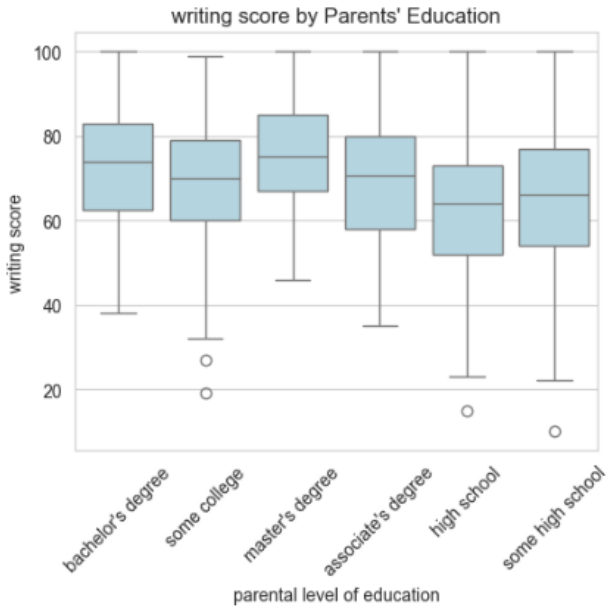
- Math improves with parental education.
- High school or less → lower scores.
- Effect smaller than reading/writing

Reading Score



- Strong link with parental education.
- Educated parents → much higher reading.
- Biggest gap across groups.

Writing Score



- Writing rises with parental education.
- Clear differences between groups.
- Similar to reading trend.

Key Insights and Recommended Actions

Key Insights:

- ❑ Higher parental education levels are associated with higher student performance across all subjects.
- ❑ Reading and writing scores are more strongly affected than math scores.
- ❑ Test preparation courses improve performance across all parental education groups.

Recommended Actions:

1. **Schools should prioritize academic support programs for students whose parents have lower educational attainment.**
2. **Test preparation courses should be made more accessible, especially for disadvantaged groups.**
3. **Reading and writing interventions may be particularly effective in reducing performance gaps.**

Project Access and Reproducibility

All materials related to this project are publicly accessible:

- ✓ ***GitHub repository (code, notebook, documentation):***
  https://github.com/hseiinshamas/StudentsPerformanceAnalysis -

- ✓ ***Cleaned dataset: included in the repository***
- ✓ ***Presentation and PDF: this document***

All analysis steps are fully documented and reproducible using the provided code and data.

LICENSE

License
This work is licensed under a **Creative Commons Attribution 4.0 International License (CC BY 4.0).**

You are free to:
•**Share** — copy and redistribute the material in any medium or format
•**Adapt** — remix, transform, and build upon the material for any purpose, even commercially
Under the following terms:
•**Attribution** — You must give appropriate credit.

Link to license:
https://creativecommons.org/licenses/by/4.0/

Students Performance Analysis

Thesis: Does Parents' Education Level Affect Students' Academic Performance?

In [1]:

```
# 1. Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (10,6)
```

In [2]:

```
# 2. Load Dataset
df = pd.read_csv('StudentsPerformance.csv')  # Change path if needed
df.head()
```

Out[2]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

-All documented on Github (the whole notebook)-